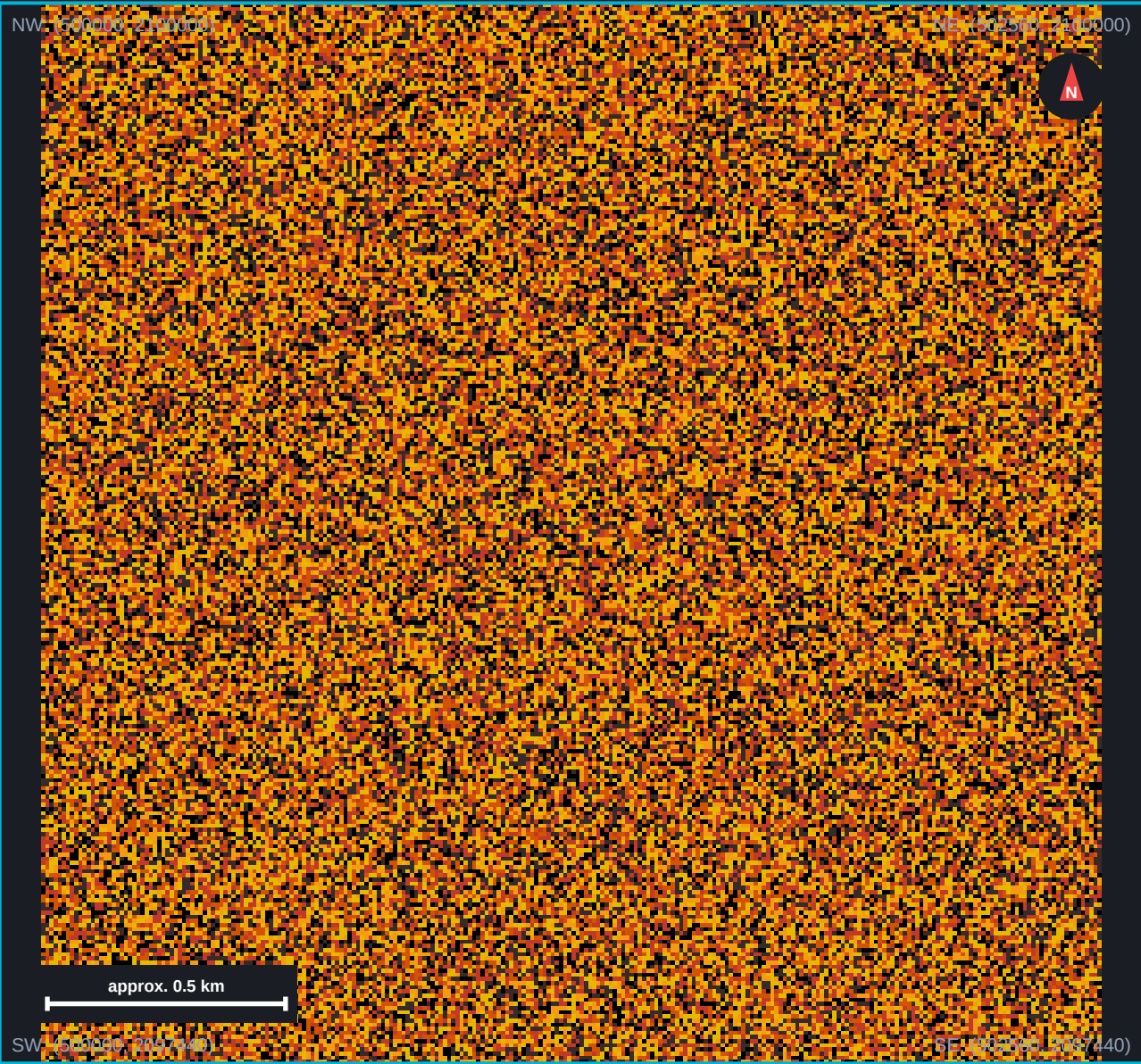


GEOSPATIAL SOIL CLASSIFICATION MAP

Source: sample_geotiff.tif | CRS: EPSG:32643 | Resolution: 10.0 m/px | Standard: USDA/FAO



SOIL CLASSIFICATION LEGEND

	Alluvial Soil Fertility: Very High Permeability: Moderate	20.0% (1.09 km ²)
	Black Soil (Vertisol) Fertility: High Permeability: Low / Impermeable	19.9% (1.09 km ²)
	Red Soil Fertility: Medium to Low Permeability: High	20.0% (1.09 km ²)
	Lateritic Soil Fertility: Low / Acidic Permeability: Moderate to High	19.8% (1.08 km ²)
	Sandy / Desert Soil Fertility: Very Low Permeability: Very High	20.3% (1.11 km ²)

PROCESSING SUMMARY

Total Mapped Pixels: 65,536
Classified Land Area: 5.45 km²
Segmentation Engine: Dual-Head PyTorch U-Net
Confidence Calibration: Entropy-Weighted Softmax
Execution Mode: Standalone Offline Local Inference